TUSHAR PATIL

Technology & Product Leader | AI-Enabled Health-Tech Expert | Cloud & Edge AI Systems Architect

Navi Mumbai, India | embedologist@gmail.com | www.linkedin.com/in/tjp24 | embedologist.in | +919819401006

EXECUTIVE SUMMARY

Technology & Product Leader with **15+ years of experience** across AI-enabled healthcare, wearables, embedded and connected-device platforms, with strong expertise in **Edge AI, LLMs, Generative AI, IoT and hardware-software integration**. Brings **6 years of LLM & SLM experience**, including deployment of AI/LLM solutions on edge environments, along with **Computer Vision, TinyML, sensors, embedded systems and cloud-edge architectures**. Proven technology leadership in building 0-to-1 products and leading cross-functional teams across hardware, embedded software and AI/ML. Patent holder in AI-based activity classification. Skilled in directing cross functional engineering teams, architecting Edge AI based Devices, API integrated and cloud-native health tech platforms, and translating emerging AI capability into regulated, market-ready products.

CORE SKILLS

AI Data & Hardware Platforms: Edge AI Integration Strategy for Medical Devices, Smart Devices Driven Digital Transformation, AI-Enabled on Device Diagnostics & Analytics, Computer Vision-Based Edge AI Systems, On Device Multimodal Bio-signal Data Integration, TinyML capable Wearable Devices, AI/ML Hardware and software Vendor & Platform Evaluation.

LLM & Edge Deployment: LLM & SLM based Edge Solutions, LLM & SLM on Device Deployment using LiteRT & PyTorch, AI/ML Deployment on customize hardware, AI Integration with Physical Devices and Multimodal Sensors

Cloud & Architecture: GCP & Google Vertex AI, AWS Practitioner, Systems & Solution Architecture, IoT & Sensor Integration, BLE/NFC Data Pipelines, Embedded Systems & Firmware.

Product Leadership: Product Vision & Roadmap, 0-to-1 Product Development & Deployment, Go-to-Market Strategy, Executive & Stakeholder Communication, Cross-functional Engineering, Technology Architecture.

Healthcare & Medical Devices: Regulatory Compliance (HIPAA, EMC), Verification & Validation (V&V), OEM/ODM & Supply Chain Partnerships, Technology Transfer & Scaling.

Delivery & Governance: Agile & Iterative Development, Program & Budget Management, Cross-Functional Team Building & Mentoring, Patents & IP Development.

PROFESSIONAL EXPERIENCE

Lead Device Engineer — Capability Architect | Dr. Reddy's Laboratories *10-2023 to 06-2026*

- Led scoping of AI integration into a multi-domain medical device roadmap, defining requirements and use cases with engineering and clinical stakeholders establishing a structured path for embedding AI capability into next-generation therapeutic devices.
- Led evaluation and architecture of AI/ML, GenAI and SLM integration for medical-device platforms, with focus on edge deployment, AI inference, hardware-software integration and physical-device use cases.
- Led a cross-functional COGS optimization on a digital therapeutic device, streamlining the BOM and coordinating global sourcing established scalable manufacturing with the world's largest device manufacturer, reducing cost and lead time.
- Evaluated AI/ML vendors, platforms, and tools against product, technical, and compliance requirements for a regulated device environment, informing platform decisions for AI-enabled device initiatives.
- Drive AI-focused digital transformation initiatives across design, devices, and engineering functions, advancing the organization's AI and digital maturity alongside core device programs.
- Own end-to-end product lifecycle management across concurrent therapeutic device programs including verification, validation, and EMC compliance delivering on schedule while maintaining regulatory readiness.

Chief Technology Officer | Oxstren Wearable Tech - Actofit *01-2015 to 09-2023*

- Built an AI/ML-enabled wearable device and health diagnostics company from zero, owning technology and product strategy end-to-end vision, roadmap, architecture, and execution scaling core product lines from 0 to 1 across multiple device categories.
- Led development and integration of AI/ML, LLM and SLM capabilities across connected wearable and health-tech platforms, including AI-based solutions and deployment in edge environments like wearable smart watches, iOS and Android mobile devices.
- Architected cloud - edge AI solutions, integrating AI models with physical devices, embedded systems, multimodal sensors and real-time data pipelines.
- Architected “Human Metabolic Health tracking & management platform” and deployed on Edge (mobile apps). Integrated wearable Bio sensors, embedded systems, cloud, and AI-ML (including Google Vertex AI-based pipelines) into unified, data-driven product experiences, improving patient engagement by 30% on the flagship AI health tech platform.

- Directed product and engineering delivery for an AI driven (Google LiteRT) mobile device based exercise analysis platform, using computer vision tracking of 32 body joints across iOS/Android with real-time voice and visual feedback. Built an early AI co-pilot style product experience shipping a real-time, camera-based coaching product for physiotherapy use cases.
- Built and mentored cross-functional teams spanning hardware, embedded software, AI/ML, and systems integration, establishing a delivery cadence that sustained product velocity through multiple release cycles.
- Established OEM/ODM partnerships and global supplier networks, scaling products from prototype to mass production while aligning the technical roadmap with go-to-market objectives.

Assistant Professor & Research Associate | DJSCOE - Mumbai University *07-2012 to 12-2014*

- Led early stage research and product discovery in BioMEMS and advanced instrumentation; mentored engineering teams and drove IP creation across multiple student-led product projects.

Embedded Application Engineer | AGD Biomedicals *07-2011 to 06-2012*

- Engineered embedded hardware architecture and firmware for diagnostic products, improving performance, reliability, and time-to-market.
- Delivered an embedded Linux-based blood cell counter with integrated cloud connectivity an early cloud-connected diagnostics product.

KEY PROJECTS

AI-Based Human Metabolism Monitoring & Recommendation Platform *36 months*

Owned product strategy and delivery for a personalized health-insights platform built on multi-source bio-signal integration (ECG, SpO2, CGM) over BLE/NFC, with Edge Ai for processing & cloud-based analytics on Google Vertex AI. Delivered a 30% improvement in patient engagement and 25% improvement in monitoring accuracy, strengthening early detection and proactive care for Human Metabolic Diseases.

AI-Base Cardiology – Small language Model: Sub-512MB Multimodal Edge Cardiology AI model | *Open Sourced* *6 months*
Architected and open-sourced cardiac SLM, an edge-native multimodal cardiology AI for Wear OS and mobile devices (<340MB RAM). Built a lightweight 1D Conformer encoder processing 25Hz PPG signals to classify 5 cardiac rhythms with 97% accuracy and <10ms latency. Integrated bio-signals via cross-attention into a 4-bit student LLM with on-device clinical RAG (1,500 guidelines), optimized for Google LiteRT (Android NNAPI). Open sourced on - https://huggingface.co/litert-community/Cardiac_micro_model_Android_Wear

Design, Development of Wearable Health Monitoring Devices *30 Months*
Designed and developed wearable smartwatches and heart rate monitors for health monitoring. Led the entire product life cycle, including ideation, design, development, ODM manufacturing, and ecosystem integration.

AI-Based Exercise Analysis Platform *24 months*
Directed product and engineering delivery for a real-time physiotherapy exercise analysis and form-correction platform, using computer-vision tracking of 32 body joints across iOS/Android with real-time voice and visual feedback an early AI co-pilot style product experience. Shipped a real-time, camera-based coaching product for physiotherapy use cases.

COGS Optimization of Digital Therapeutic Device *24 months*
Led a cross-functional redesign to reduce manufacturing cost without sacrificing quality, through BOM streamlining and global sourcing coordination. Established scalable manufacturing with the world's largest device manufacturer, reducing cost and lead time for Edge AI driven prescription Medical devices.

Digital Therapeutic SaMD & SiMD System's Technical Management & Regulatory Compliance *10 months*
Owned end-to-end product lifecycle engineering and technical management for digital therapeutic SaMD & SiMD devices, leading verification and validation (V&V), EMC testing, and FDA design controls readiness in compliance with ISO 13485 and IEC 60601

Wearable ECG & SpO2 Measurement Device *18 months*
Owned product design and go-to-market for a medical-grade TinyML capable wearable ECG/SpO2 device (chest bands, smartwatches) with a secure BLE pipeline to a companion app.

Human Body Composition Analyzer *24 months*
Led development of a bioimpedance body composition analyser with CDMO partners in China, improving measurement reliability by 20% and cutting production cost by 15%.

5-Part Blood Cell Counter *10 months*
Delivered an embedded Linux-based 5-part blood cell counter from hardware architecture, Firmware development through manufacturing readiness.

PATENTS & PUBLICATIONS

“Training of Classifiers for Identifying Activities Based on Motion Data” — WIPO Patent WO2018134646. ML-based system for classifying user activity from motion data using trained classifiers, foundational IP underpinning AI-driven wearable products.

EDUCATION & CERTIFICATIONS

M.Tech, Instrumentation & Control (Specialization Biomedical Instrumentation)
COEP Technological University, Pune (06-2011)
Grade - 8.35/10

Bachelor of Engineering, Biomedical Engineering
Mumbai University (08-2009)
Grade – 75.10%